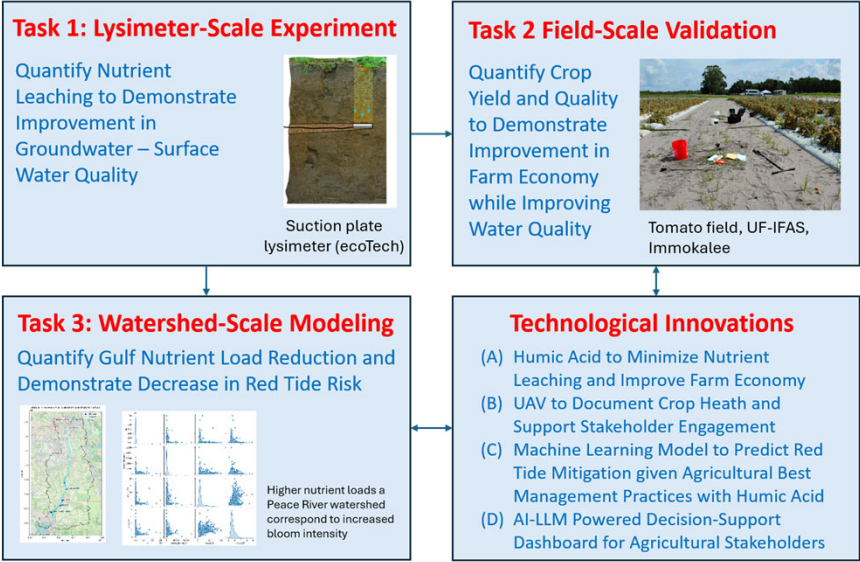


FGCU-UF Planning Workshop - Sep 8, 2026

- Schedule review
- Update: Orders for Task 1 and Task 2
- QAPP discussion and Site Selection

	2026		2027				2028				2029	
Task/Deliverable	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
Task 1. Lysimeter Experiment to Quantify Nutrient Leaching Reduction (2 seasons)												
Task 2. Field Scale Validation with UAV Imagery to Demonstrate Economic Feasibility (1 season)												
Task 3a. Watershed Model to Quantify Nutrient Load Reduction												
Task 3b. Machine Learning Model to Predict Red Tide Mitigation given Agricultural BMPs												
Task 3c LLM-Powered Decision-Support Dashboard for Agricultural Stakeholders												
Reporting												

Humic Acid for reducing nutrient loads and HAB risks



Updates – Orders for Task 1 and Task 2

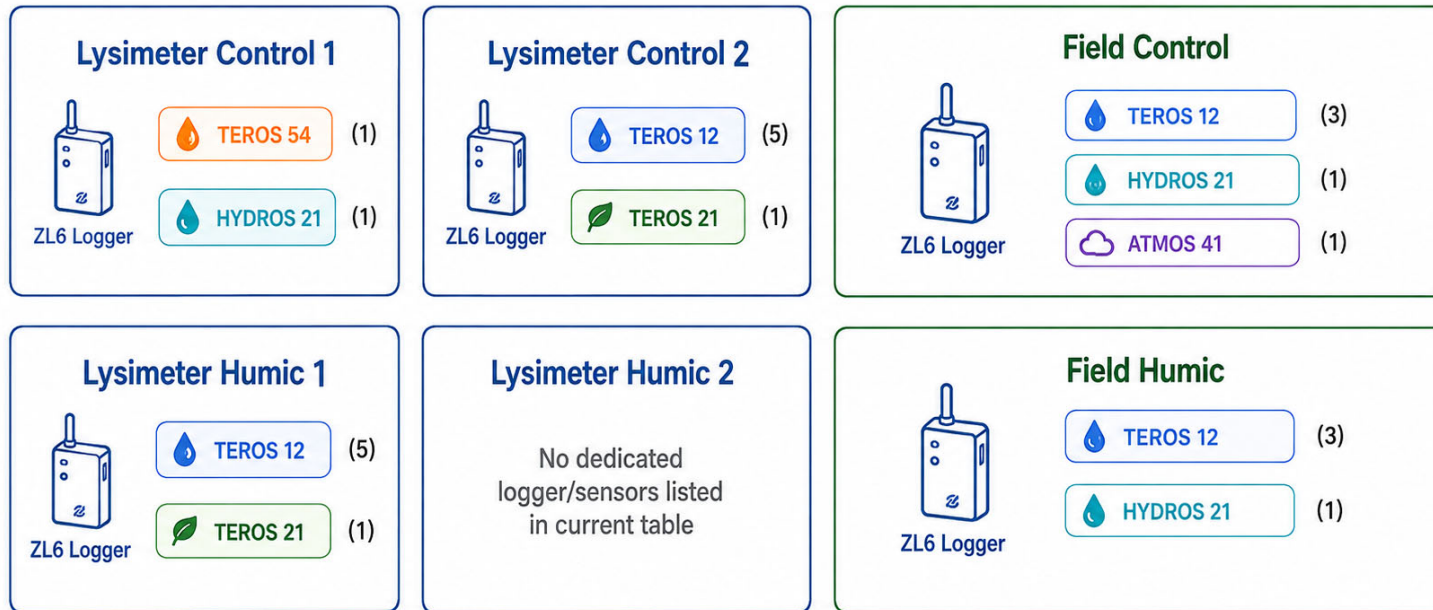
FGCU

- We ordered lysimeter (~\$12,000, delivery date to be confirmed)
- We are placing order for soil moisture sensors, soil water potential sensors, water level loggers from METER (~\$5,000)
- We are working on ordering completion materials for 5 shallow GW monitoring wells, and 2 water level loggers from In-Situ (~\$3,000)
- We are working on ordering Humic Acid

UF

- Identify materials and supply needed to start experiments in Task 1 and Task 2 in Jan growing seasons (starting with the list on Teams)

Sensor and Logger Layout



	MODEL	Needed_Q	Existing_Q	Order_Q	Unit Price	Total Price	Comments
Soil Moisture	TEROS12	16	5	11	271	2,981	Lysimeter (5 units x 2 exp) + Field (3 sets x 2 exp)
Soil Moisture	TEROS 54	1	1	0	1,076	-	
Soil Tension	TEROS21	2	1	1	288	288	Lysimeter (1 unit x 2 exp)
GW Level	HYDROS21	3	1	2	692	1,384	Will get two additional InSitu sensors
Weather Station	ATMOS41	1	1	0	2,620	-	
Data Logger	ZL6	5	5	0	766	-	Field Control: TEROS12(3)+HYDRO21(1)+ATMOS(1) Field Humic: TEROS12(3)+HYDROS(1) Lysimeter Control1: TEROS 54(1) + HYDROS (1) Lysimeter Control2: TEROS12(5) + TEROS 21 (1) Lysimeter Humic1: TEROS12(5) + TEROS 21 (1)
Telemetry	ZENTRA	0	0	0	209	-	Cloud standard plan (12 month)
						-	
					Cost	4,653	

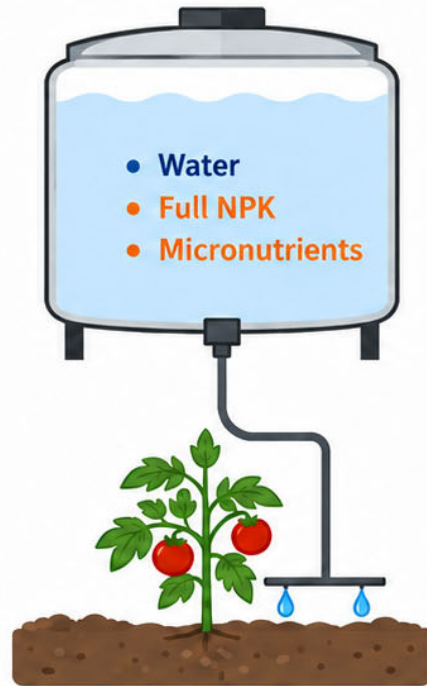
Four Experimental Irrigation Treatments

CONTROL 1



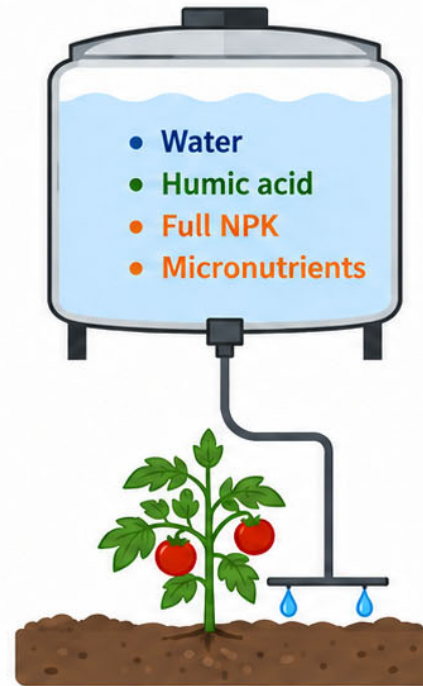
No agriculture

CONTROL 2



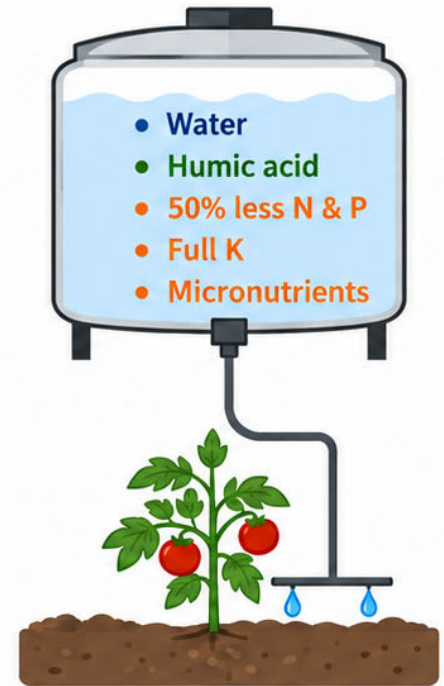
Pre-plant: Compost/manure

TREATMENT 1



Pre-plant: Compost/manure

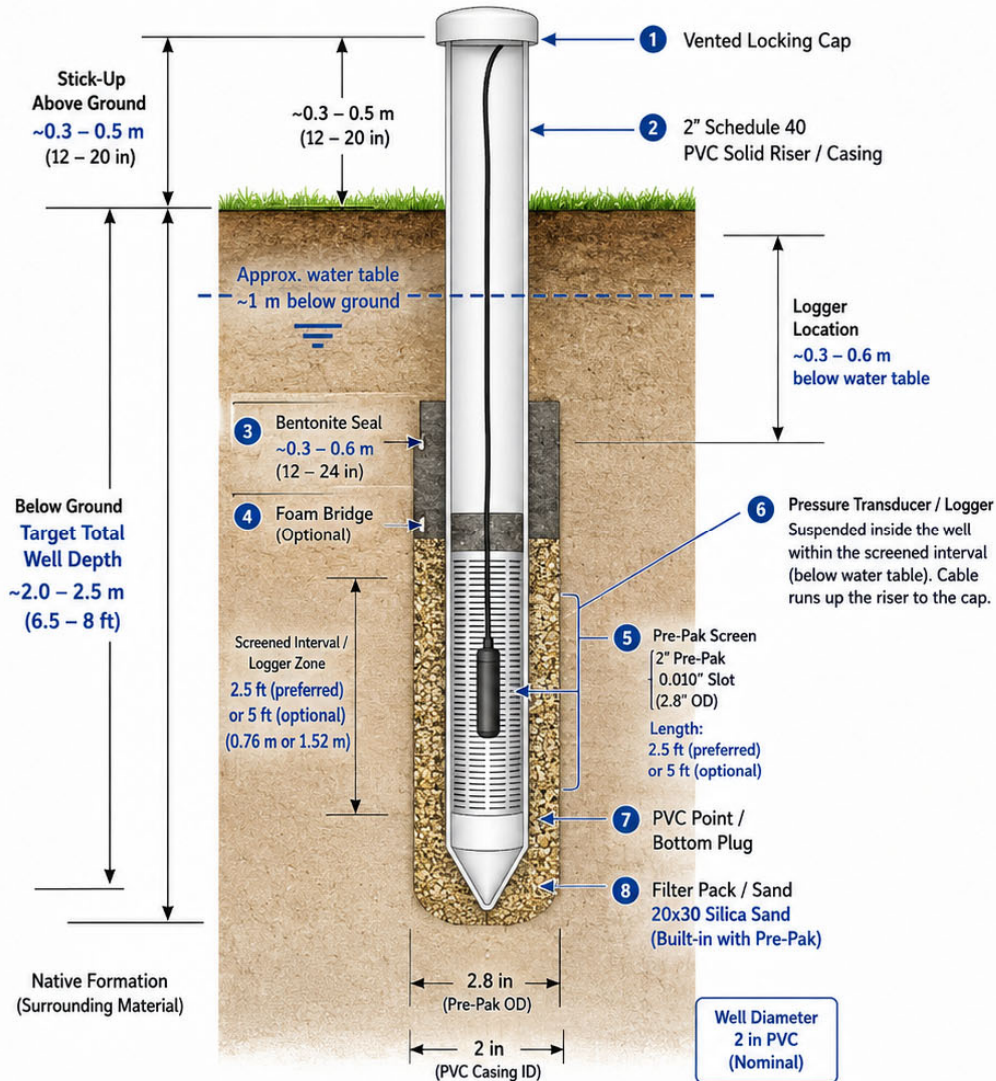
TREATMENT 2



Pre-plant: Compost/manure

All tanks supply irrigation water. Compost/manure is applied to the soil before planting—not added to the irrigation tank.

ATLANTIC SUPPLY PRE-PAK MONITORING WELL – DETAILED DIMENSIONED DIAGRAM



HOW ATLANTIC QUOTE ITEMS ARE USED

1 Locking Plug / Cap (02VENTEDCAP)	Closes the top of the well, keeps out debris. Vented cap allows the pressure transducer (TROLL/HYDROS) to read true water level.
2 Solid PVC Casing (02PVC40 .20)**	Riser pipe from top of screen to ground surface. Provides a place to hang the logger and protects the well.
3 Bentonite Chips (EC700, 50 lb)**	Hydrated in the annulus above the screen to form a low-permeability seal. Prevents surface water from traveling down the outside of the casing.
4 Foam Bridge (FB2)	Temporary foam barrier placed on top of the screen before adding bentonite to the seal material does not fall into the screen and clog it.
5 Bentonite Quick Seal (02BWQS)	Pre-assembled/contained bentonite sealing section. Another way to create the seal. (Often used instead of loose bentonite chips in some installations.)
6 Pre-Pak Screen (02PEP2X5X010)**	The screened interval of the well. 2" Pre-Pak with 0.010" slot, 20x30 silica sand, 2.8" OD. Water enters the well through this screen.
7 PVC Point / Bottom Plug (02PVCPOINT2)	Closes the bottom of the well and allows the well to be driven/installed without collapsing.
8 Filter Pack / Sand (20x30 Silica Sand)**	Surrounds the screen (built in with Pre-Pak) and may also be added around the lower portion during installation to ensure good hydraulic connection.

** Quantities in the old quote (e.g., 40) were for a different project (many wells). For your 5 shallow wells you will need only enough materials for five wells.

INSTALLATION NOTES (SUMMARY)

- Auger/advance borehole to target depth.
- Place Pre-Pak screen section.
- (Optional) Place foam bridge on top of screen.
- Hydrate/insert bentonite seal above screen.
- Add sand to base (if not pre-packed all the way).
- Install solid PVC riser to surface.
- Glue on locking/vented cap.
- Suspend logger; verify depth (~0.3–0.6 m below water table).
- Backfill and finish flush around casing.

QAPP discussion and site selection

Pending issue in QAPP including site selection for the Lysimeter and Field experiments to setup workflow and collect samples before Jan growing season

- **Review QAPP:** include your inputs with more focus to Task 2 related sections
- **Select sites:** consider uniform sandy soil, irrigation access, **drainage and flooding conditions**
- **Experimental design designs:** plot size, randomization, replicates, cultivar (tomato variety that will be planted), and treatment rate and timing (how often humic acid and fertilizer will be applied)
- **Equipment installation:** suction-plate depth (tomato root depth?), sensors, well, logger, weather station and irrigation checks.
- **Sample collection:** Collection and preservation before sending them to lab, confirm sampling frequency (the plan is twice weekly)
- **Baseline sampling:** Before actual experiment (January). This might help us to evaluate previous land management variability.

From workshop decision to January field launch

	QAPP	Orders and Installations	Sampling
Sep – Week 1-2	- Select site (Task 1 and 2) - Design Experiments - QAPP to EPA		
Sep – Week 3-4		- Order well casing, humic acid materials and supplies (Task 1&2)	- Decide on lab - Sampling plan
Oct - Week 1-2	- Finalize QAPP	- Install well casing - Install sensors	- Student hiring and training
Oct - Week 3-4		- Install lysimeter	
Nov - Week 1-2		- Irrigation tanks for exp	- Baseline sampling
Nov - Week 3-4		- Test setup	- Data analysis and finalize sampling plan
Dec - Week 1-2			
Dec – Week 3-4			- Begin monitoring
Jan – Week 1-2			- Begin Task 1 & 2